

DogeCoin/BTC – LTC/BTC lead lag

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Background: Litecoin is a cryptocurrency marketed as the silver to bitcoins gold. Litecoin is also mined on Scrypt which acts as an auxiliary proof of work for Dogecoin. Since Litecoin is the parent currency of Scrypt all miners must use simultaneously mine Dogecoin and Litecoin. The way Litecoin mining works is that Litecoin targets a set rate of cryptocurrency mined each hour adjusting difficulty every 2016 blocks or 12600 Litecoin depending on current hash power (mining power).

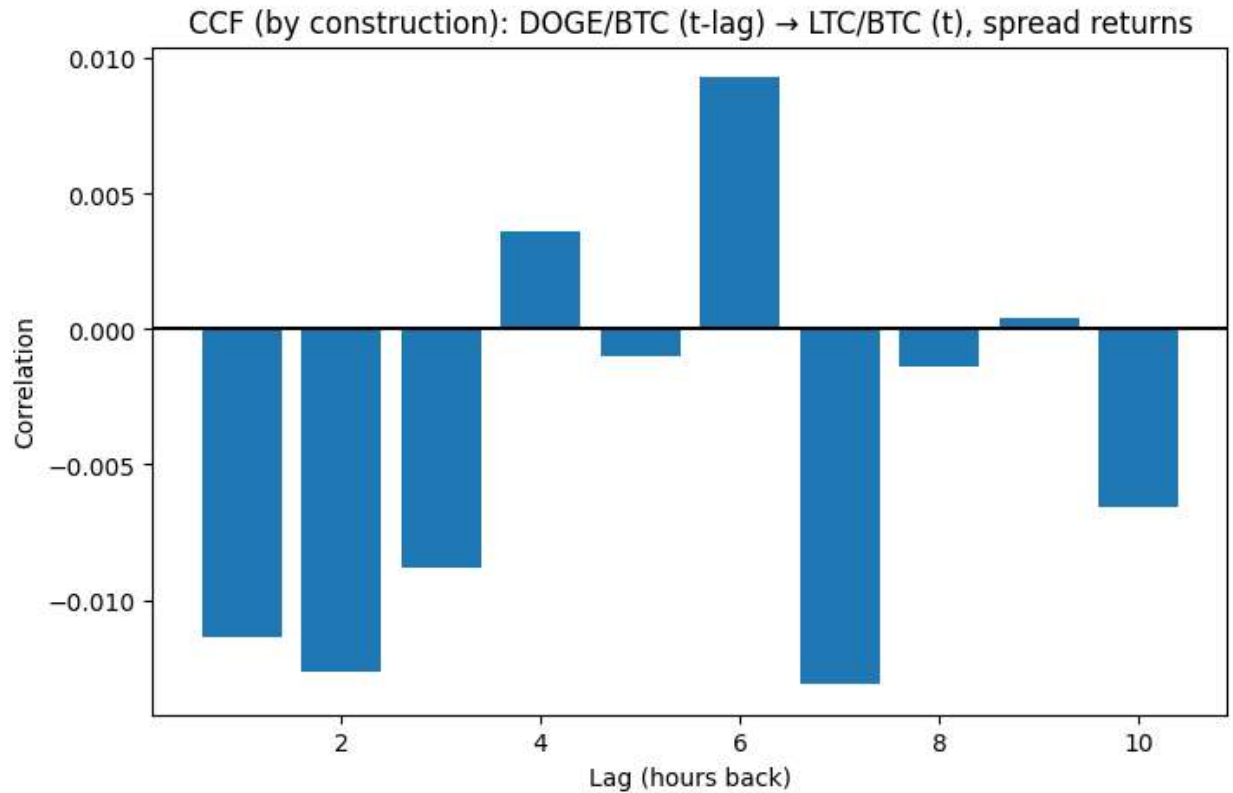
Thesis: I attempt to use Dogecoin to determine when there are more Litecoin miners and more hash power. Since Dogecoin miners must mine Litecoin, when prices of Dogecoin increase, Litecoin miners also likely increase as miners attempt to seek higher Dogecoin prices. Because difficulty for Litecoin is adjusted sporadically, the increase in miners should temporarily increase the rate of supply. Though small, this would cause some supply side pressure to lower the price of Litecoin relative to Bitcoin.

From a demand side, I believe hedging to bitcoin removed a lot of the sentiment. This should remove a lot of the short-term pressures derived from demand, thereby isolating the supply side thesis. It also removes the necessity of taking a view of the crypto market.

EDA:

I chose to use hourly data and extended my training period to include larger jumps in doge and see its stability relative to this. I chose to start use 2022 – 2024 as my training data and 2025-jan 2026 as my testing data.

I ran tests first on the data unfiltered. I started with a CCF to test if there was any lagged effect.



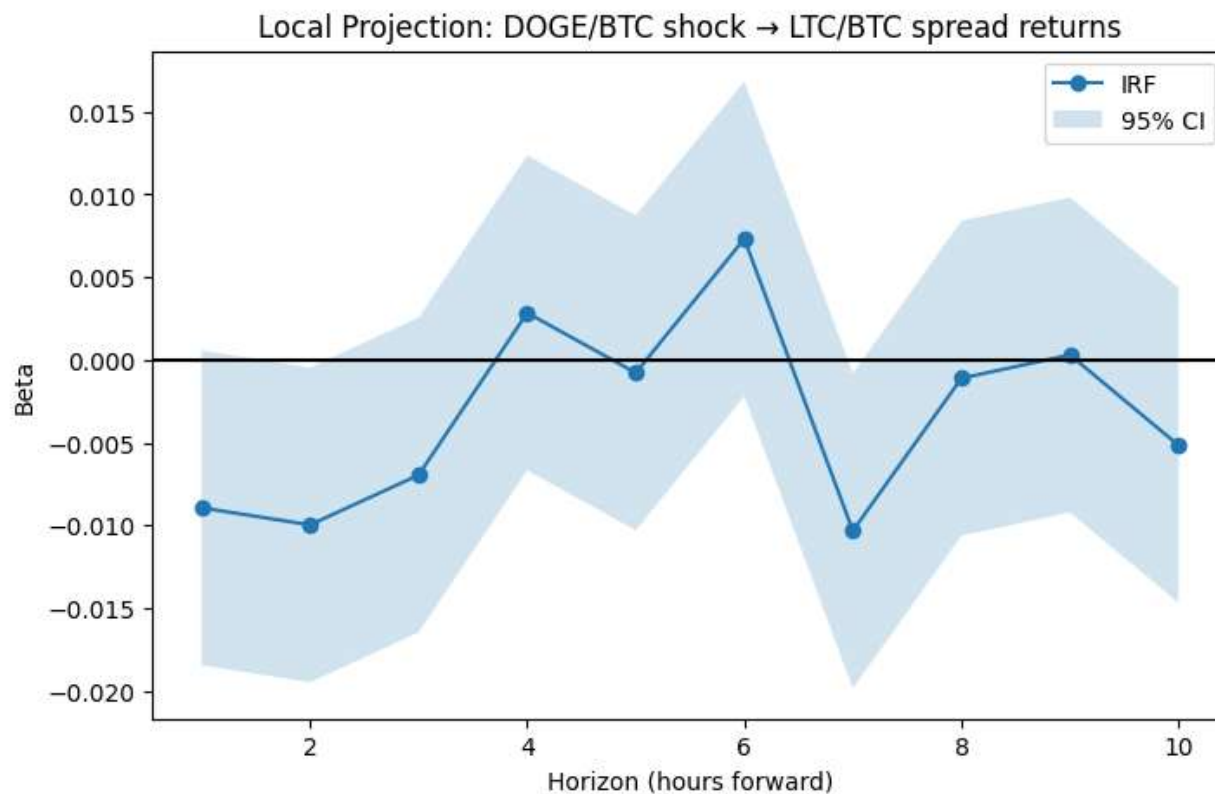
- 1 -0.011351
- 2 -0.012662
- 3 -0.008833

Based on these results it seemed effects of the supply pressure may be realized very quickly. This aligns with my thesis as difficulty readjustment would close the supply pressure as well as potential sentiment shifts and autocorrelating dampening the effects later. The small correlation also aligns with thesis as the effects should be small due to the small, expected increase in supply. From this I ran a Granger Causality test, OLS test, and local projection test for lags 1-10 to test to test these results

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Granger results:

- 1 $p = 0.065668$
- 2 $p = 0.040051$
- 3 $p = 0.152093$



Horizon 1:

beta(shock) = -8.934134×10^{-3}

std err = 4.853508×10^{-3}

t-stat = -1.841

p-value = 0.065668

Horizon 2:

beta(shock) = -9.965931×10^{-3}

std err = 4.853555×10^{-3}

t-stat = -2.053

p-value = 0.040051

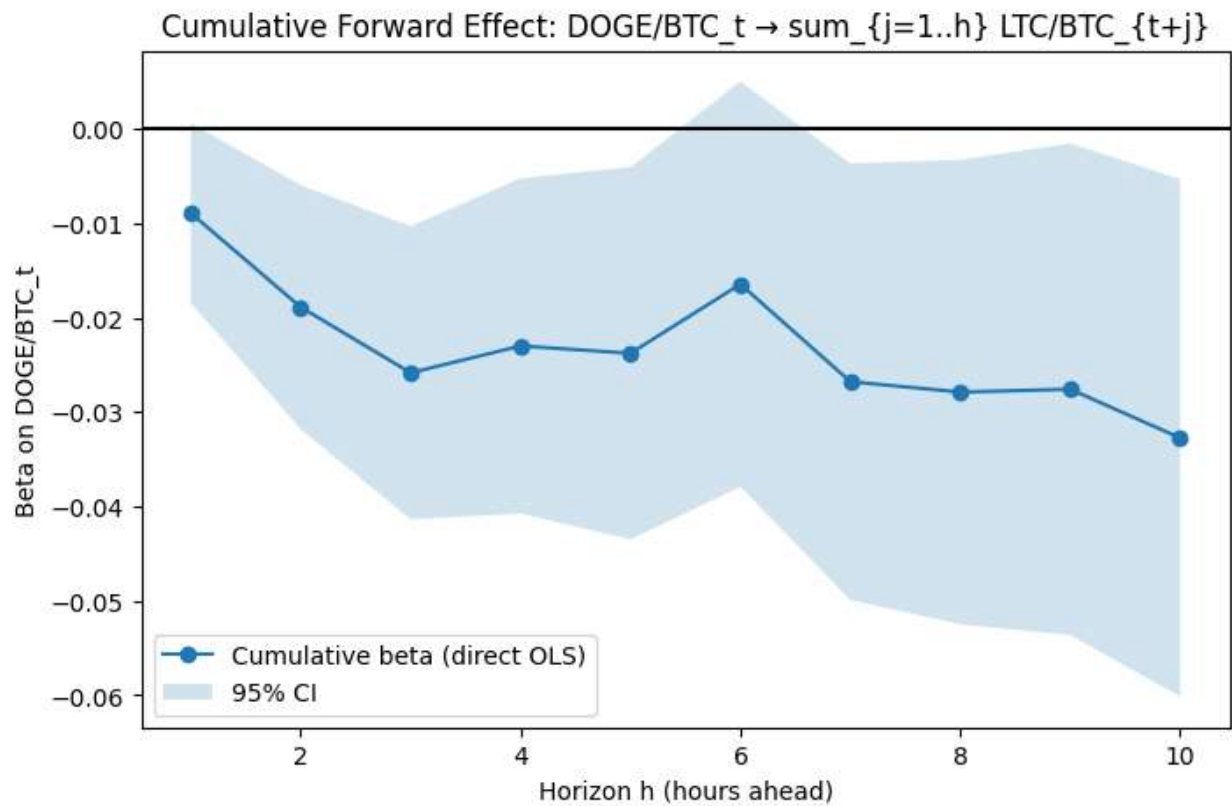
Horizon 3:

beta(shock) = -6.951707×10^{-3}

std err = 4.853797×10^{-3}

t-stat = -1.432

p-value = 0.152093



Horizon 1 (cumulative 1..1 hours ahead):

n obs = 26296
beta = -8.934133e-03
std err = 4.853415e-03
t-stat = -1.841
p-value = 0.065663

Horizon 2 (cumulative 1..2 hours ahead):

n obs = 26296
beta = -1.890004e-02
std err = 6.591240e-03
t-stat = -2.867
p-value = 0.004141

Horizon 3 (cumulative 1.3 hours ahead):

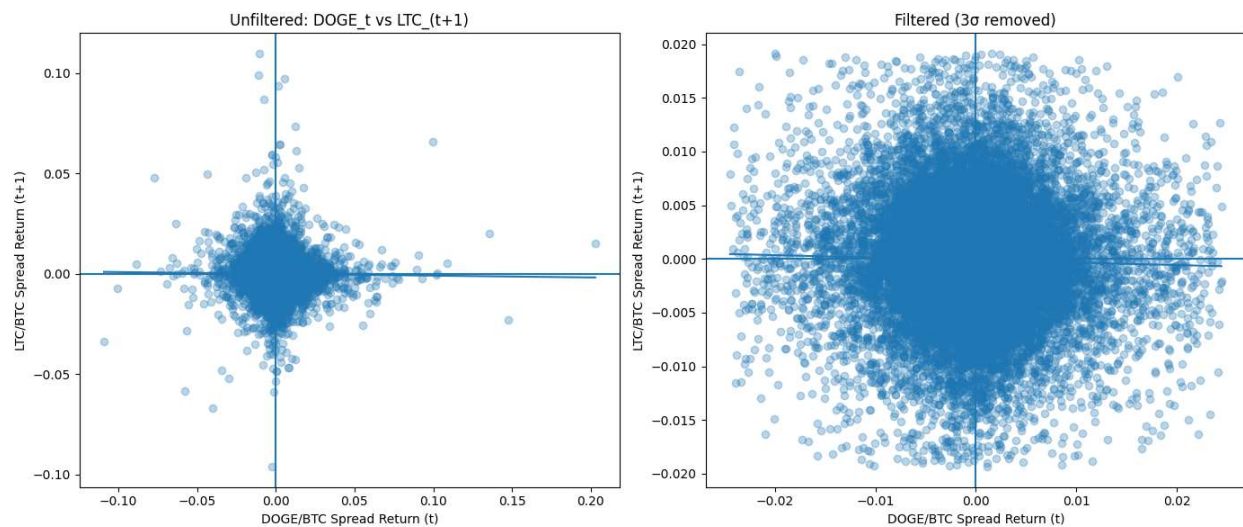
beta = -2.585176e-02

std err = 7.921247e-03

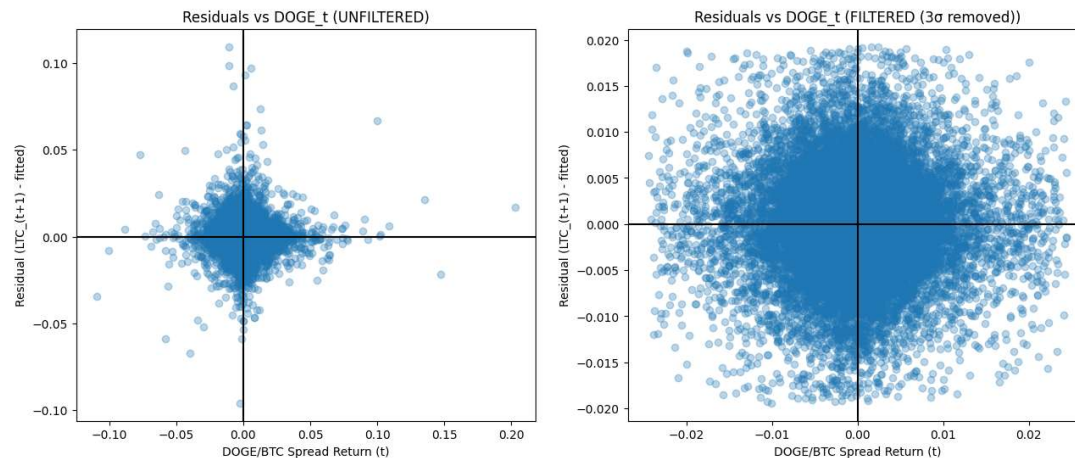
t-stat = -3.264

p-value = 0.001101

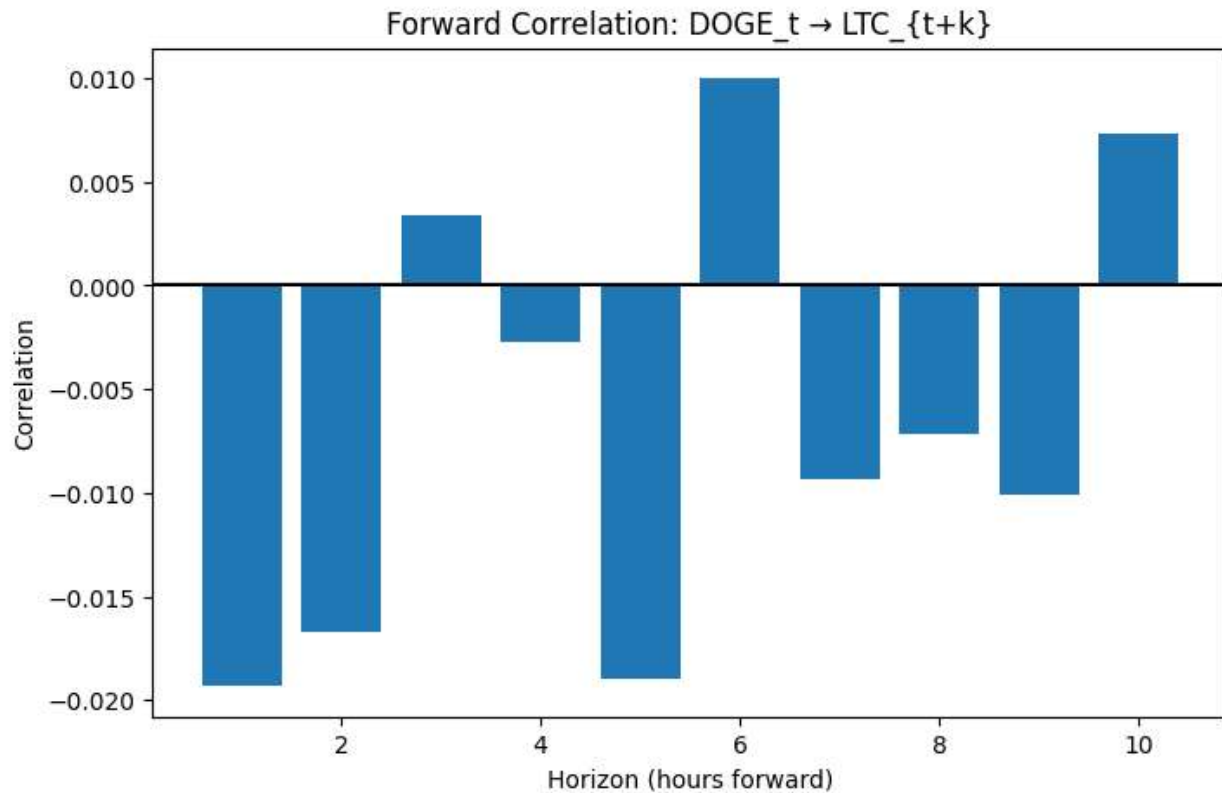
These results seemed significant with near significant p-values for all tests at a lag of 1 and significant at 2. These results show that the correlation between doge/btc and ltc/btc is significantly negative for short term lags of 1 and 2 lags. These tests, however, all assume some linearity which may be offset by outlier points. To see whether they had outliers I plotted out one hour lagged results.



Clearly the filtered results looked better in terms of outlier points. I also wanted to test if linear was the best so I graphed out the residuals.



The graph does not appear to have any obvious patterns for both so the linear assumption did seem to hold. I reran the tests on the filtered data.

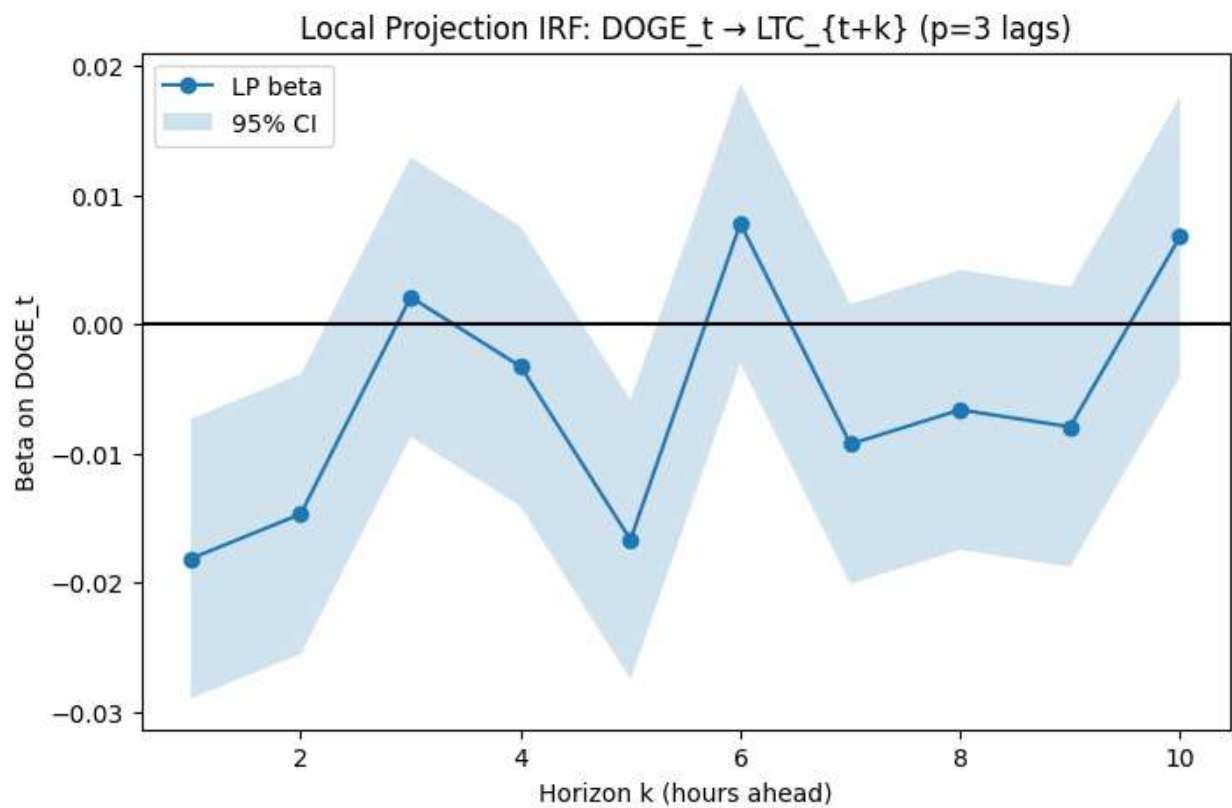


1 -0.019319
 2 -0.016671
 3 0.003370

Gragner result:

1 $p = 0.017262$

2 $p = 0.014849$



Horizon 1:

beta = -1.814155e-02

std err = 5.521642e-03

t-stat = -3.286

p-value = 0.001019

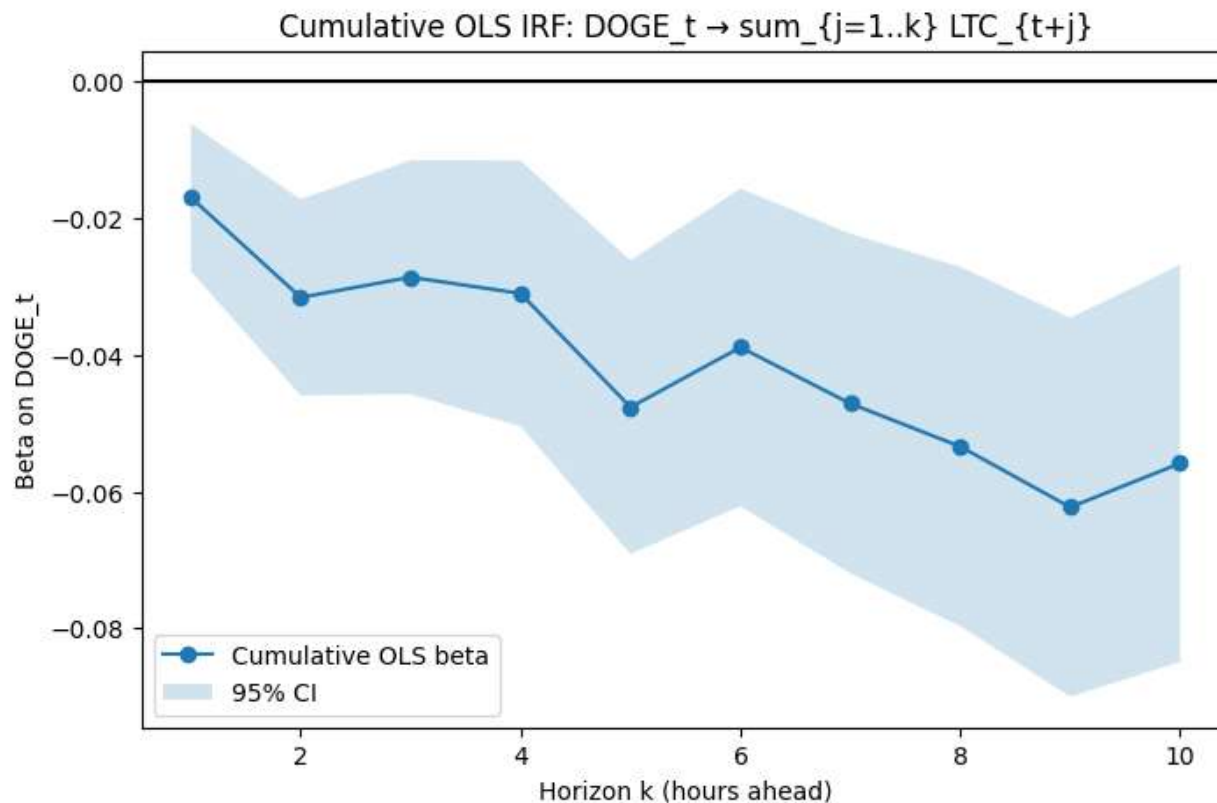
Horizon 2:

beta = -1.468840e-02

std err = 5.521576e-03

t-stat = -2.660

p-value = 0.007815



Horizon 1 (cumulative 1..1 hours ahead):

beta = -1.696210e-02

std err = 5.497379e-03

t-stat = -3.085

p-value = 0.002034

R-sq = 0.000373

Horizon 2 (cumulative 1..2 hours ahead):

beta = -3.159970e-02

std err = 7.329376e-03

t-stat = -4.311

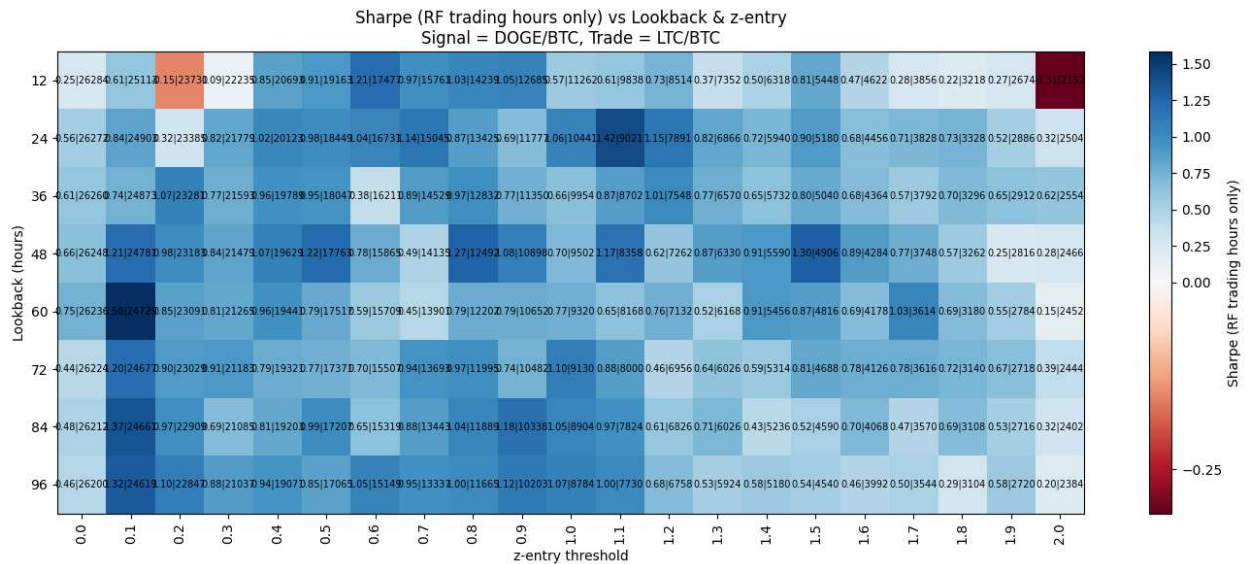
p-value = 0.000016

R-sq = 0.000728

The results from data filtering are still significant for the 1 and 2 lag if not more significant. I believe this testing is enough for me to start back testing.

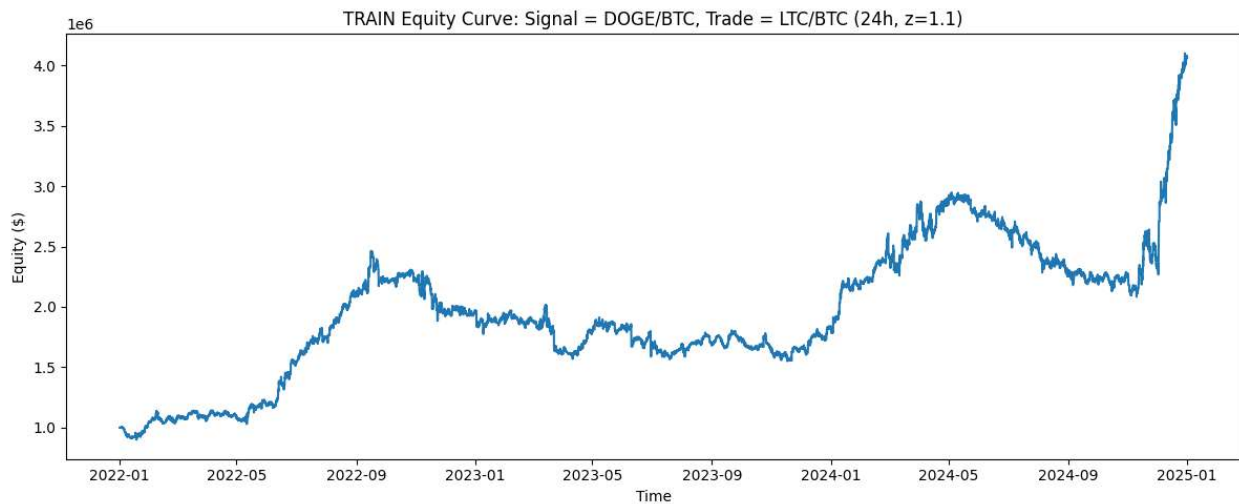
Back testing

I start by creating a heatmap optimizing on lookback period and z-score. I chose not to filter as I wanted to remain liable to high z-score moves in dogecoin. Due to extremely low spreads on Kraken and no fee for large volume trades, I used a 0.1 bip transaction cost. I also chose to use a 2 hour holding period and a lag of one due to the high statistical significance with the first two hours.



Due to robustness I chose the 1.1 z-score entry and the 24 hour lookback period

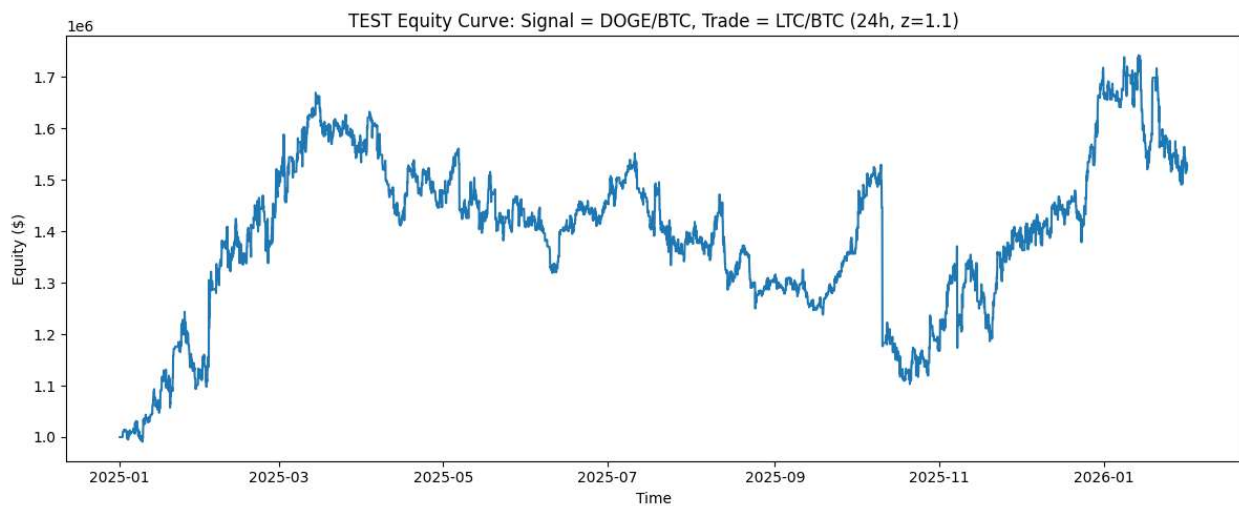
Training



Cumulative return: 3.0641
Annualized return: 0.5954
Annualized volatility: 0.3662
Sharpe (RF trading hours only): 1.4217

Sharpe (RF all hours): 1.3499
Total trades: 4511
Losses to TC (\$): 354260.49
Max drawdown: -0.3703
Days in position: 376

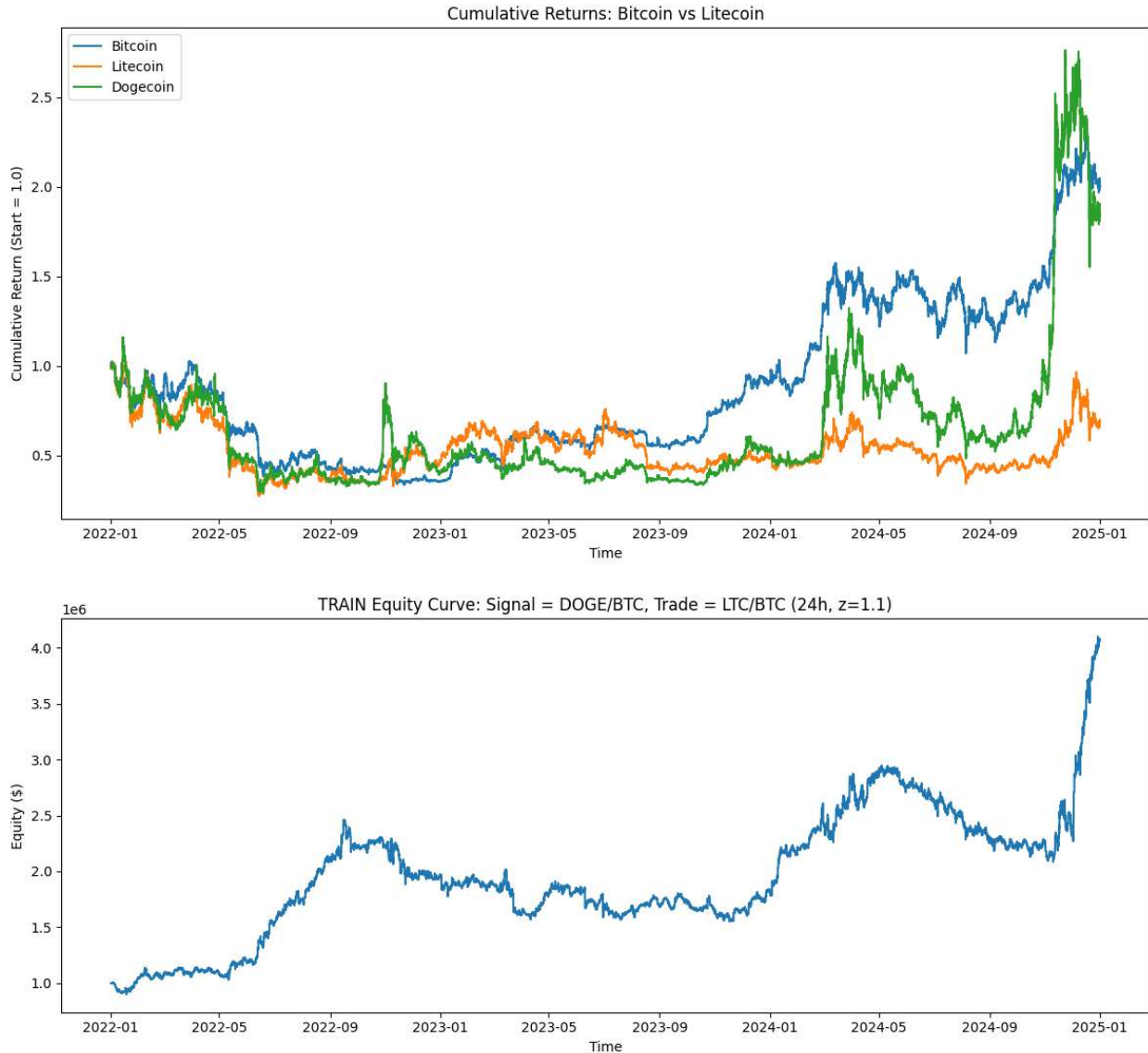
Testing



Cumulative return: 0.5280
Annualized return: 0.4782
Annualized volatility: 0.5151
Sharpe (RF trading hours only): 0.9916
Sharpe (RF all hours): 0.9425
Total trades: 1746
Losses to TC (\$): 97325.88
Max drawdown: -0.3393
Days in position: 146

Editing Signal

I went further into my signal and analyzed why there might be some regime like influence. For this I graphs out my cumulative returns for all three coins and compared them to the returns of the strategy over my training period.

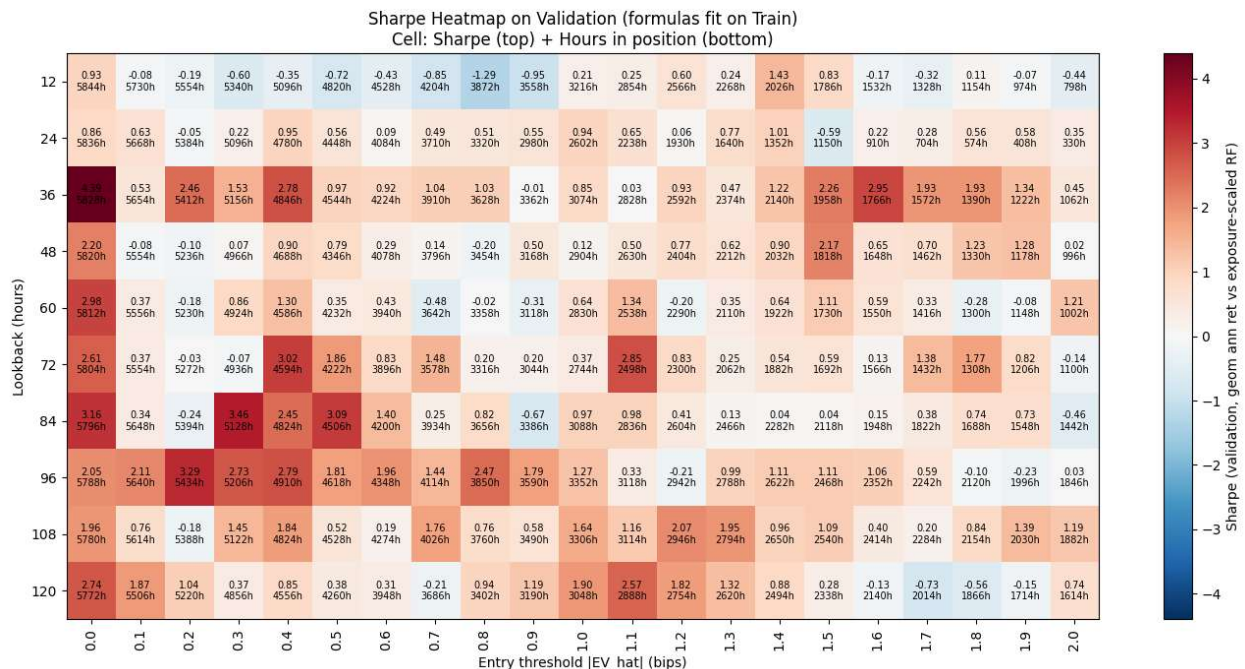


It seems that a lot of my profit is made when beta of Dogecoin changes relative to bitcoin. Due to this and the good results of my current signal, I decided to develop an OLS (multivariate linear regression model) for the expected value of a hold of 2 hours (in bps) relative to a rolling beta over a lookback period and my previous signal, z-score of change of $\log(\text{dog/bitcoin})$ based on a lookback period. I decided to also divide my time into a training, testing and validation period to ensure that I don't overfit to one period. I calculated the EV formulas based on the training period (2022-2023), optimized over the validation period (2024), and tested over the testing period (2025-Jan 2026).

Based on the training period, the following values were derived for different lookback periods:

$[12h] \text{ EV} = -2.68461 + (0.130541) \cdot Z + (1.54619) \cdot B + \varepsilon \quad | n=17498$
 $[24h] \text{ EV} = -2.70166 + (0.0695685) \cdot Z + (1.60587) \cdot B + \varepsilon \quad | n=17486$
 $[36h] \text{ EV} = -3.8551 + (-0.0440404) \cdot Z + (2.55964) \cdot B + \varepsilon \quad | n=17474$
 $[48h] \text{ EV} = -4.02098 + (-0.00681971) \cdot Z + (2.72753) \cdot B + \varepsilon \quad | n=17462$
 $[60h] \text{ EV} = -4.12513 + (0.0374469) \cdot Z + (2.83981) \cdot B + \varepsilon \quad | n=17450$
 $[72h] \text{ EV} = -4.29578 + (0.0580054) \cdot Z + (3.00754) \cdot B + \varepsilon \quad | n=17438$
 $[84h] \text{ EV} = -5.1946 + (0.110963) \cdot Z + (3.78212) \cdot B + \varepsilon \quad | n=17426$
 $[96h] \text{ EV} = -6.29683 + (0.0779568) \cdot Z + (4.73838) \cdot B + \varepsilon \quad | n=17414$
 $[108h] \text{ EV} = -6.34555 + (0.0901) \cdot Z + (4.7972) \cdot B + \varepsilon \quad | n=17402$
 $[120h] \text{ EV} = -5.79973 + (0.0312175) \cdot Z + (4.37257) \cdot B + \varepsilon \quad | n=17390$

With these formulas I chose to optimize for lookback period and entry expected value. I optimized on sharpe where rfr was only subtracted from hours in position.



Due to robustness I chose to test on 0.5 bps and 84 hour lookback period.

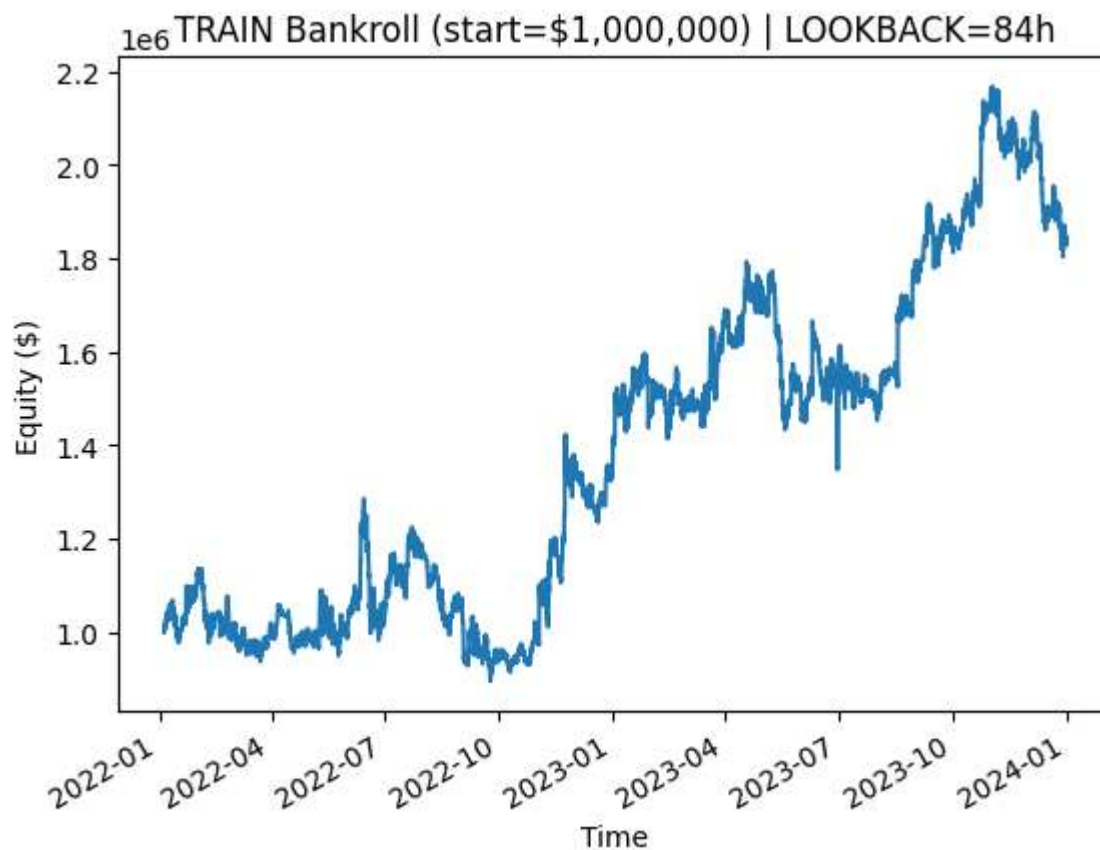
===== TRAIN (84h, EV_entry=0.5 bips) BACKTEST =====

Cumulative return: 0.8356

Annualized return: 0.3571

Annualized volatility: 0.4288

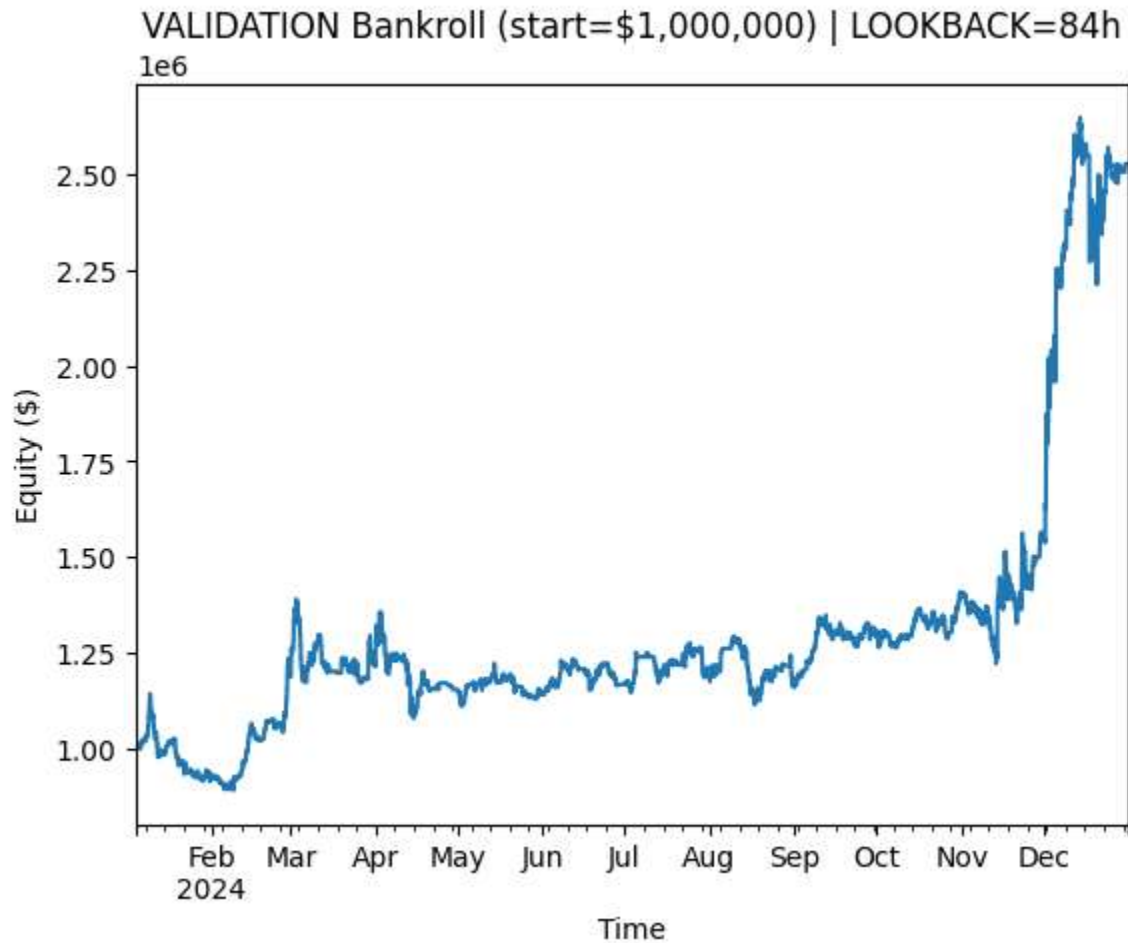
Sharpe (RF all hours): 0.7395 (rf_eff=0.0400)
Sharpe (RF scaled by exposure): 0.7779 (rf_eff=0.0235)
Exposure (time in position): 0.5935
Total trades: 5171
Losses to TC (\$): 51710.00
Max drawdown: -0.3030
Days in position: 431



===== VALIDATION (84h, EV_entry=0.5 bips) BACKTEST =====

Cumulative return: 1.5283
Annualized return: 1.5453
Annualized volatility: 0.4940
Sharpe (RF all hours): 3.0469 (rf_eff=0.0400)
Sharpe (RF scaled by exposure): 3.0863 (rf_eff=0.0205)
Exposure (time in position): 0.5181
Total trades: 2253

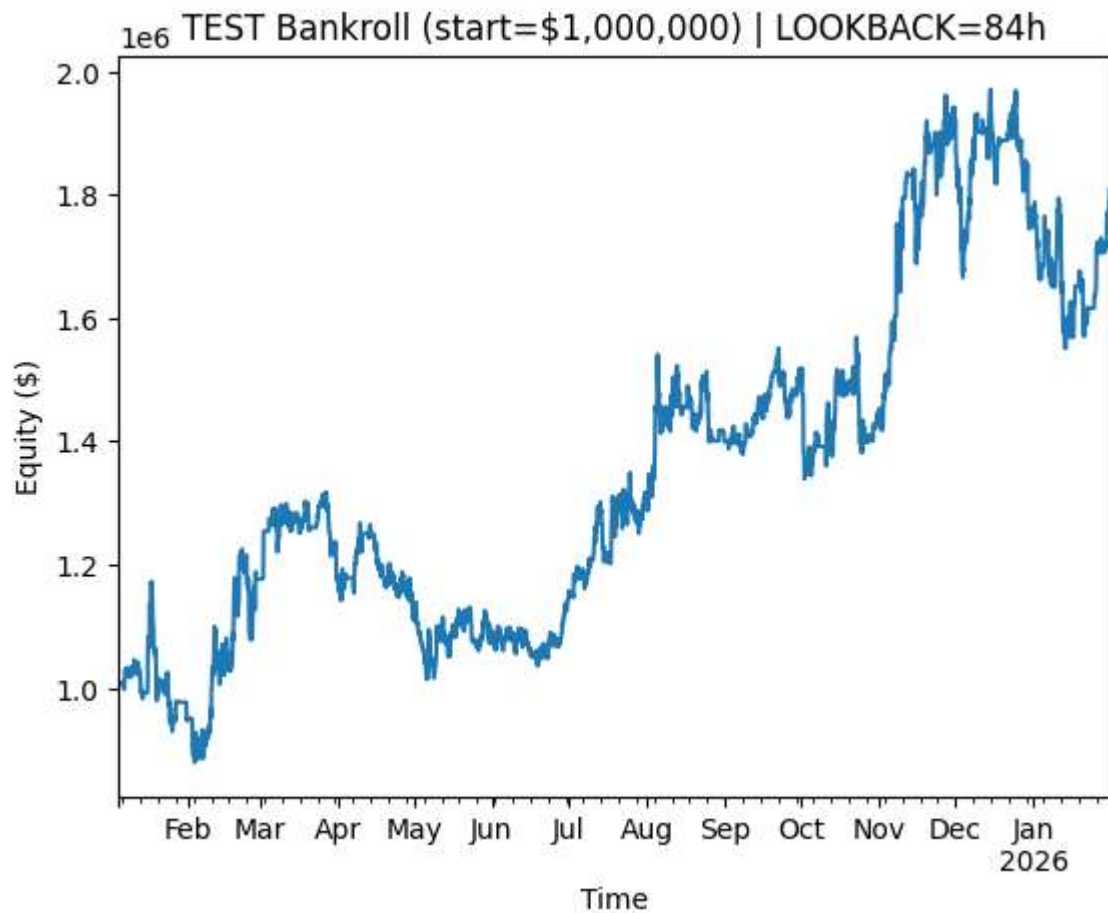
Losses to TC (\$): 22530.00
Max drawdown: -0.2243
Days in position: 188



===== TEST (84h, EV_entry=0.5 bips) BACKTEST =====

Cumulative return: 0.7636
Annualized return: 0.6951
Annualized volatility: 0.4959
Sharpe (RF all hours): 1.3211 (rf_eff=0.0400)
Sharpe (RF scaled by exposure): 1.3603 (rf_eff=0.0205)
Exposure (time in position): 0.5186
Total trades: 2442
Losses to TC (\$): 24420.00
Max drawdown: -0.2499

Days in position: 204



Based on the results I seem to have a pretty good edge trading my strategy. I am however a bit worried about the quick fall off outside the region of good sharpe on the heatmap. Due to this I chose to test on the 84 hour lookback and 0.7 sharpe point.

===== TRAIN (84h, EV_entry=0.7 bips) BACKTEST =====

Cumulative return: 0.3337

Annualized return: 0.1558

Annualized volatility: 0.4059

Sharpe (RF all hours): 0.2852 (rf_eff=0.0400)

Sharpe (RF scaled by exposure): 0.3296 (rf_eff=0.0220)

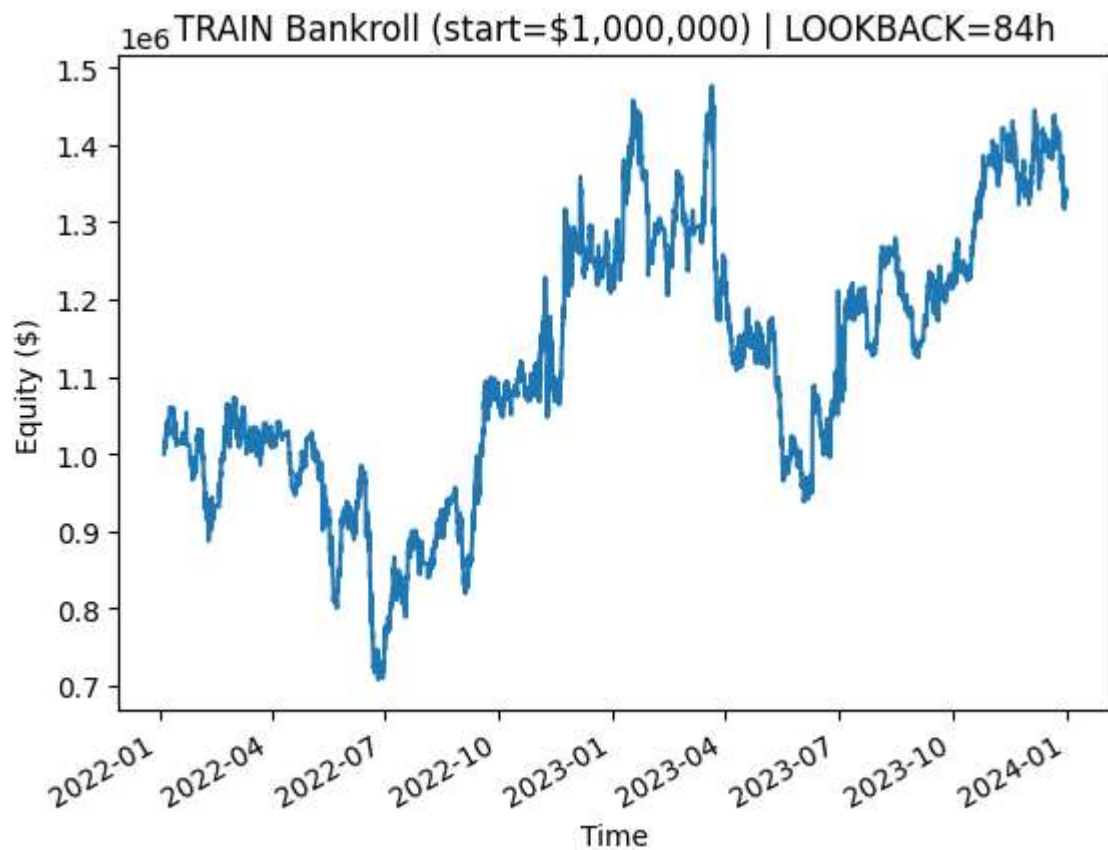
Exposure (time in position): 0.5542

Total trades: 4829

Losses to TC (\$): 48290.00

Max drawdown: -0.3642

Days in position: 402



===== VALIDATION (84h, EV_entry=0.7 bips) BACKTEST =====

Cumulative return: 0.1304

Annualized return: 0.1314

Annualized volatility: 0.4502

Sharpe (RF all hours): 0.2030 (rf_eff=0.0400)

Sharpe (RF scaled by exposure): 0.2520 (rf_eff=0.0179)

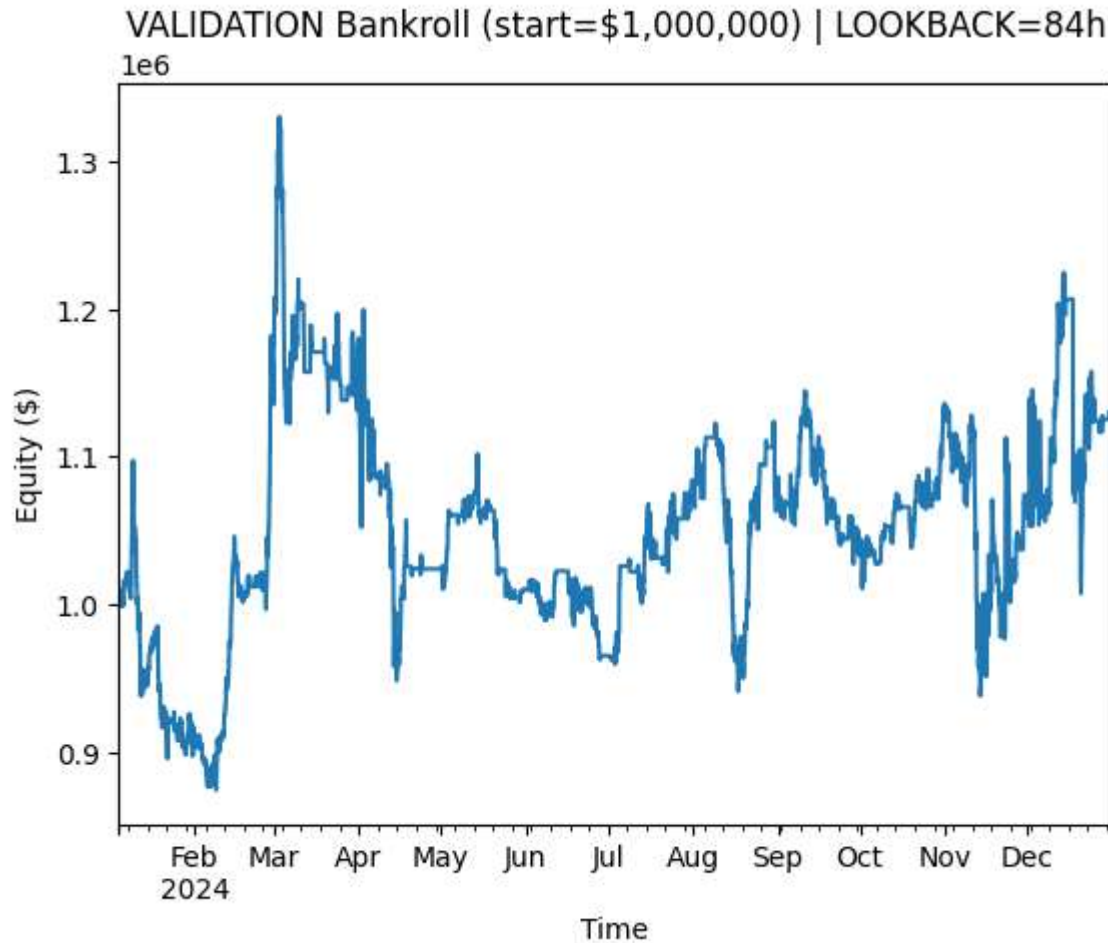
Exposure (time in position): 0.4523

Total trades: 1967

Losses to TC (\$): 19670.00

Max drawdown: -0.2951

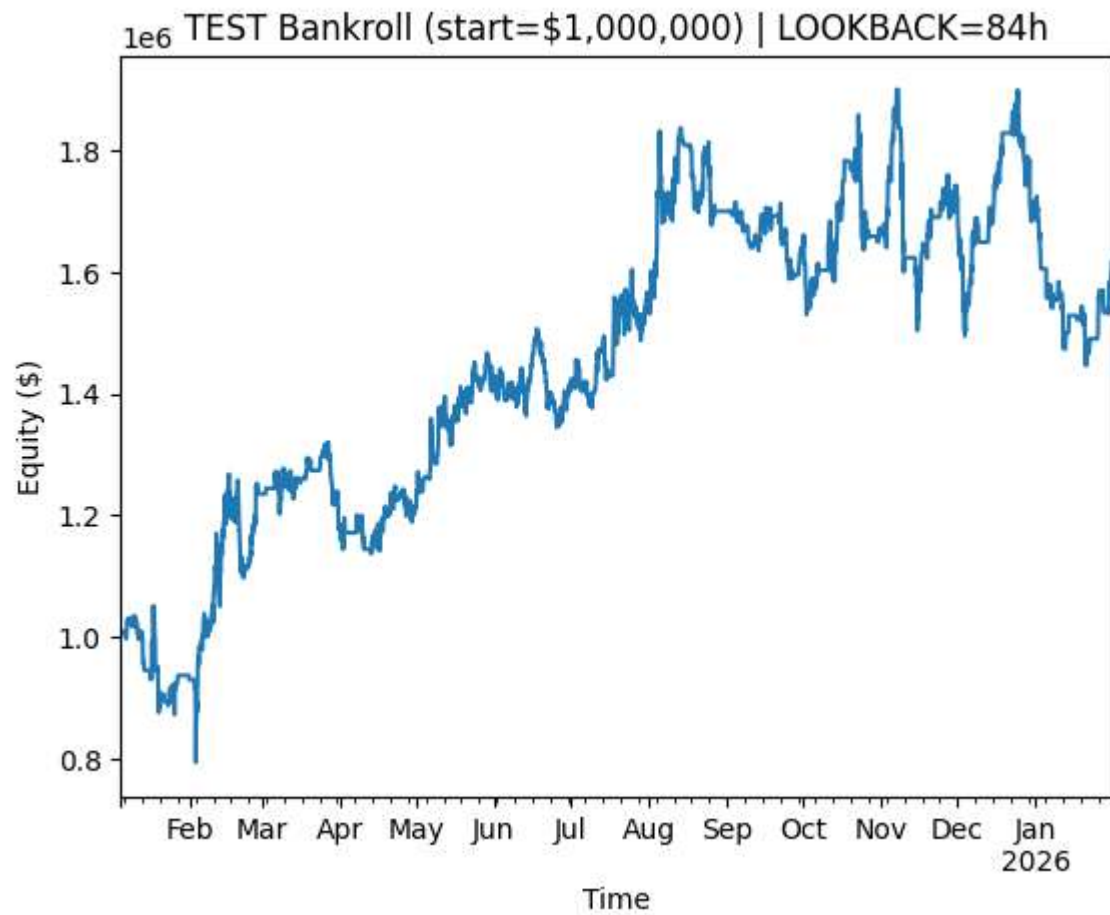
Days in position: 164



===== TEST (84h, EV_entry=0.7 bips) BACKTEST =====

Cumulative return: 0.5757
Annualized return: 0.5265
Annualized volatility: 0.4618
Sharpe (RF all hours): 1.0535 (rf_eff=0.0400)
Sharpe (RF scaled by exposure): 1.1017 (rf_eff=0.0177)
Exposure (time in position): 0.4486
Total trades: 2112
Losses to TC (\$): 21120.00
Max drawdown: -0.2447

Days in position: 176



Due to the positive results outside of the validation period, I believe there might be variance issues affecting the training period. Due to this, I believe the edge does indeed hold.